

NICOLAS REED

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EDUCATION

University of California, Berkeley

GPA: 3.63/4.00

B.S. Electrical Engineering & Computer Sciences (EECS)

Expected May 2027

Planning to pursue an M.S.; seeking a Summer 2027 internship

Coursework: Tapeout, Bringup*, Digital Design and Integrated Circuits (with ASIC Lab), Computer Architecture, Operating Systems, Compilers*, Signals and Circuits, Data Structures, Computer Networking, Data Science

EXPERIENCE

Apple

Cupertino, CA

SoC Design Verification Intern

May 2026 – August 2026

- Optimized embedded C calibration firmware, cutting image size by over 10% while preserving execution latency.
- Designed a UVM testbench with surrounding hardware models to verify a microcontroller and its firmware.

UC Berkeley SLICE Lab

Berkeley, CA

Undergraduate Research Assistant - advised by Professor Sophia Shao

May 2025 – Present

- Profiled RISC-V BOOM using Top-Down Microarchitecture Analysis counters to identify workload bottlenecks.
- Contributed to discovery-agent infrastructure for evolutionary hardware design-space exploration.

UC Berkeley Electrical Engineering & Computer Sciences (EECS)

Berkeley, CA

Head Teaching Assistant

June 2024 – May 2026

- Manage a 50+ member course staff supporting 1,200+ students across C, RISC-V, and Python coursework.
- Ranked 1st among staff in helpfulness and teaching effectiveness.
- Author weekly homework and lab assignments, and maintain course infrastructure using GitHub and Docker.
- Lead online forum engagement for the course, answering 1,500+ student questions per semester.

Sandisk

Milpitas, CA

Software Development Intern

May 2025 – August 2025

- Reconciled CMDB and inventory data for 40,000+ virtual machines with Pandas to plan a cloud migration.
- Built a LlamaIndex AI assistant to streamline license and contract workflows across 10,000+ applications.

Micross Components - Silicon Turnkey Solutions

Milpitas, CA

Engineering Intern

July 2023 – August 2023

- Conducted mechanical and PCB testing on 1,000+ devices, root-causing compromised semiconductors.

PROJECTS

TSMC 16nm Physical AI Tapeout - Chisel, Chipyard, Hammer

December 2025 – May 2026

- Owned RTL implementation of the matrix engine for an Apple-mentored SoC tapeout.
- Coordinated block-level interfaces and system connectivity during integration of the NPU accelerator.
- Developed bare-metal functional tests, stress kernels, and VLA workloads for full-chip verification.
- Validated the design by running 130+ Atlas tests on VCU118 and Alveo U250 FPGAs, both bare and via FireSim.

5-Stage Pipelined RISC-V CPU with I/D Caches - Verilog

October 2025 – December 2025

- Apple NSI (New Silicon Initiative) Design Contest Winner: delivered best ASIC performance and area.
- Boosted throughput and frequency by refining hazard control and forwarding, reducing stall cycles.
- Achieved single-cycle read hits and two-cycle write hits with efficient write-back logic.

Pintos Operating System - C

August 2025 – December 2025

- Extended core OS support for process control and system calls, handling invalid user memory safely.
- Added multithreading and synchronization primitives, enforcing priority scheduling to prevent inversion.
- Created UNIX-style file system features including caching, file growth, and subdirectories to improve I/O.

Scheme Interpreter - Python

April 2024 – May 2024

- Implemented an interpreter for a subset of Lisp (Scheme), supporting core expressions and syntax.
- Applied semantic and lexical analysis techniques that machines use to evaluate and execute code.

TECHNICAL SKILLS

Languages: Verilog, SystemVerilog, C, Python, RISC-V, Chisel, Scala, Java, x86, SQL, MATLAB

Tools: Git, Cadence (Indago, Innovus), Synopsys (VCS, Verdi), UVM, GDB, Pandas, NumPy, Docker, LTSpice